

47 • `SELECT StudentID, FirstName, LastName, EnrollmentDate FROM Students;`

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
Wrap Cell Content: 				
	StudentID	FirstName	LastName	EnrollmentDate
▶	1	John	Doe	2022-08-01
	2	Jane	Smith	2021-08-01
	3	Charlie	Brown	2023-01-15
*	NULL	NULL	NULL	NULL

```
51  ## 2. Retrieve students who enrolled after 2022.
52 • SELECT StudentID, FirstName, LastName, EnrollmentDate
53     FROM Students
54     WHERE EnrollmentDate > '2022-12-31';
```

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
	StudentID	FirstName	LastName	EnrollmentDate
*	NULL	NULL	NULL	NULL

```
56  ## 3. Retrieve courses offered by the Mathematics department with a limit of 5 courses.
57 • SELECT T1.CourseName, T2.DepartmentName
58     FROM Courses AS T1
59     JOIN Departments AS T2 ON T1.DepartmentID = T2.DepartmentID
60     WHERE T2.DepartmentName = 'Mathematics'
61     LIMIT 5;
62
```

Result Grid		
Filter Rows:		
Export: 		
Wrap Cell Content: 		
	CourseName	DepartmentName
▶	Data Structures	Mathematics

```
63  ## 4. Get the number of students enrolled in each course, filtering for courses with more than 5 students.
64 • SELECT T2.CourseName, COUNT(T1.StudentID) AS TotalEnrollments
65     FROM Enrollments AS T1
66     JOIN Courses AS T2 ON T1.CourseID = T2.CourseID
67     GROUP BY T2.CourseName
68     HAVING COUNT(T1.StudentID) > 5;
69
```

Result Grid		
Filter Rows:		
Export: 		
Wrap Cell Content: 		
	CourseName	TotalEnrollments

```
70  ## 7. Calculate the average number of credits for all courses.
71 • SELECT AVG(Credits) AS AverageCourseCredits FROM Courses;
72
```

Result Grid		
Filter Rows:		
Export: 		
Wrap Cell Content: 		
	AverageCourseCredits	
▶	3.5000	

73 ## 8. Find the maximum salary of instructors in the Computer Science department.

74 • ALTER TABLE Instructors ADD Salary DECIMAL(10,2);

75 • UPDATE Instructors SET Salary = 80000 WHERE InstructorID = 1;

76 • UPDATE Instructors SET Salary = 70000 WHERE InstructorID = 2;

77 • SELECT MAX(T1.Salary) AS MaxCSInstructorSalary

78 FROM Instructors AS T1 JOIN Departments AS T2 ON T1.DepartmentID = T2.DepartmentID

79 WHERE T2.DepartmentName = 'Computer Science';

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	MaxCSInstructorSalary
▶	80000.00

81 ## 9. Count the number of students enrolled in each department.

82 • SELECT T3.DepartmentName, COUNT(DISTINCT T1.StudentID) AS StudentsInDepartment

83 FROM Enrollments AS T1 JOIN Courses AS T2 ON T1.CourseID = T2.CourseID JOIN Departments AS T3 ON T2.DepartmentID = T3.DepartmentID

84 GROUP BY T3.DepartmentName;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	DepartmentName	StudentsInDepartment
▶	Computer Science	1
	Mathematics	1

86 ## 5. Find students who are enrolled in both Introduction to SQL and Data Structures.

87 • SELECT T1.StudentID, T1.FirstName, T1.LastName FROM Students AS T1

88 INNER JOIN Enrollments AS E1 ON T1.StudentID = E1.StudentID

89 INNER JOIN Courses AS C1 ON E1.CourseID = C1.CourseID AND C1.CourseName = 'Introduction to SQL'

90 INNER JOIN Enrollments AS E2 ON T1.StudentID = E2.StudentID

91 INNER JOIN Courses AS C2 ON E2.CourseID = C2.CourseID AND C2.CourseName = 'Data Structures';

92

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	StudentID	FirstName	LastName
▶	1	John	Doe
	2	Jane	Smith

93 ## 6. Find students who are either enrolled in Introduction to SQL or Data Structures. (UNION Logic)

94 • SELECT T1.StudentID, T1.FirstName, T1.LastName

95 FROM Students AS T1 JOIN Enrollments AS T2 ON T1.StudentID = T2.StudentID JOIN Courses AS T3 ON T2.CourseID = T3.CourseID

96 WHERE T3.CourseName = 'Introduction to SQL'

97 UNION

98 SELECT T1.StudentID, T1.FirstName, T1.LastName

99 FROM Students AS T1 JOIN Enrollments AS T2 ON T1.StudentID = T2.StudentID JOIN Courses AS T3 ON T2.CourseID = T3.CourseID

100 WHERE T3.CourseName = 'Data Structures';

101

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	StudentID	FirstName	LastName
▶	1	John	Doe
	2	Jane	Smith

102 ## 10. INNER JOIN: Retrieve students and their corresponding courses.

103 • SELECT T1.FirstName, T1.LastName AS StudentName, T2.CourseName

104 FROM Students AS T1 INNER JOIN Enrollments AS T3 ON T1.StudentID = T3.StudentID

105 INNER JOIN Courses AS T2 ON T3.CourseID = T2.CourseID;

106

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	FirstName	StudentName	CourseName
▶	John	Doe	Introduction to SQL
	Jane	Smith	Data Structures

107 ## 11. LEFT JOIN: Retrieve all students and their corresponding courses, if any.

```
108 • SELECT T1.FirstName, T1.LastName AS StudentName, T2.CourseName
109 FROM Students AS T1 LEFT JOIN Enrollments AS T3 ON T1.StudentID = T3.StudentID
110 LEFT JOIN Courses AS T2 ON T3.CourseID = T2.CourseID;
111
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	FirstName	StudentName	CourseName
▶	John	Doe	Introduction to SQL
	Jane	Smith	Data Structures

112 ## 12. Subquery: Find students enrolled in courses that have more than 10 students.

```
113 • SELECT T1.StudentID, T1.FirstName, T1.LastName FROM Students AS T1
114 WHERE T1.StudentID IN ( SELECT StudentID FROM Enrollments WHERE CourseID IN ( SELECT CourseID FROM Enrollments GROUP BY CourseID HAVING COUNT(StudentID) > 10 ));
115
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	StudentID	FirstName	LastName
▶	1000	John	Doe

116 ## 13. Extract the year from the EnrollmentDate of students.

```
117 • SELECT FirstName, LastName, EnrollmentDate, YEAR(EnrollmentDate) AS EnrollmentYear FROM Students;
118
```

119 ## 14. Concatenate the instructor's first and last name.

Result Grid Filter Rows: Export: Wrap Cell Content:

	FirstName	LastName	EnrollmentDate	EnrollmentYear
▶	John	Doe	2022-08-01	2022
	Jane	Smith	2021-08-01	2021

119 ## 14. Concatenate the instructor's first and last name.

```
120 • SELECT InstructorID, CONCAT(FirstName, ' ', LastName) AS FullName FROM Instructors;
121
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	InstructorID	FullName
▶	1	Alice Johnson
	2	Bob Lee

122 ## 15. Calculate the running total of students enrolled in courses. (Window Function)

```
123 • SELECT T1.EnrollmentID, T1.StudentID, T1.EnrollmentDate, COUNT(T1.StudentID)
124 OVER (ORDER BY T1.EnrollmentDate, T1.EnrollmentID) AS RunningTotalEnrollments FROM Enrollments AS T1;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	EnrollmentID	StudentID	EnrollmentDate	RunningTotalEnrollments
▶	2	2	2021-08-01	1
	1	1	2022-08-01	2

126 ## 16. Label students as 'Senior' or 'Junior' based on their year of enrollment. (CASE Expression)

```
127 • SELECT StudentID, FirstName, LastName, EnrollmentDate,
128 CASE WHEN (YEAR(CURDATE()) - YEAR(EnrollmentDate)) >= 4 THEN 'Senior' ELSE 'Junior' END AS StudentStatus FROM Students;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	StudentID	FirstName	LastName	EnrollmentDate	StudentStatus
▶	1	John	Doe	2022-08-01	Junior
	2	Jane	Smith	2021-08-01	Senior

Export recordset to an external file